MAT 201 - MATHEMATICAL METHODS I

LECTURE 7: PARTIAL DERIVATIVES AND APPLICATIONS

LECTURE C24-1-7 :: MON., JAN. 27, 2025 8:00 AM (WAT) - FRI., JAN. 31, 2025 6:00 PM (WAT)

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OUTLINE

- Introduction
- Derivatives of a Function of Two Variables
- Functions of More Than Two Variables
- Higher-Order Partial Derivatives
- Selected Applications
- Conclusion

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INTRODUCTION

- **Overview:**
  - Partial derivatives are a fundamental concept in calculus.
  - They extend the idea of derivatives to functions of multiple variables.
  - Essential for understanding the rate of change in multivariable functions.

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INTRODUCTION

- **Notation:**
  - The partial derivative of a function of several variables with respect to one variable is its derivative concerning that variable, with the other variables held constant.
  - For a function $f(x, y)$, the partial derivative with respect to x is denoted as $\frac{\partial f}{\partial x}$.
  - Similarily, the partial derivative with respect to y is denoted as $\frac{\partial f}{\partial y}$.

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INTRODUCTION

- **Definition:**

$$\frac{\partial f}{\partial x} = f_x(x,y) = \lim_{h\to 0} \frac{f(x+h,y) - f(x,y)}{h} \tag{1}$$

$$\frac{\partial f}{\partial y} = f_y(x,y) = \lim_{h\to 0} \frac{f(x,y+k) - f(x,y)}{k} \tag{2}$$

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INTRODUCTION

- **Significance:**
  - Multivariable Optimization: Crucial in optimizing functions of multiple variables, such as maximizing profit in economics or minimizing energy in physics.
  - Gradient: Partial derivatives form the gradient vector, which indicates the direction of steepest ascent of a function.
  - Taylor Series: Used to approximate functions around a point in multivariable calculus.
  - Partial Differential Equations: Essential in solving equations involving multiple variables, such as heat equations or wave equations.

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INTRODUCTION

- **Geometric Interpretation:**
  - Partial derivatives provide insights into the slope of tangent lines in the x and y directions.
  - The directional derivative in the x direction is $\frac{\partial f}{\partial x}$, and in the y direction is $\frac{\partial f}{\partial y}$.
  - These derivatives help understand how the function changes concerning each input variable.

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DERIVATIVES OF A FUNCTION OF TWO VARIABLES

- **Example 1:** $f(x,y) = x^2 - 3xy + 2y^2 - 4x + 5y - 12$

$$\begin{aligned}
\frac{\partial f}{\partial x} = f_x(x,y) &= \lim_{h\to 0} \frac{f(x+h,y) - f(x,y)}{h} && (3)\\
&= \lim_{h\to 0} \frac{2xh + h^2 - 3hy - 4h}{h} && (4)\\
&= \lim_{h\to 0} \frac{h(2x + h - 3y - 4)}{h} = \lim_{h\to 0}(2x + h - 3y - 4) && (5)\\
&= 2x - 3y - 4 && (6)
\end{aligned}$$

DERIVATIVES OF A FUNCTION OF TWO VARIABLES

- **Example 1 [Cont.]:** $f(x,y) = x^2 - 3xy + 2y^2 - 4x + 5y - 12$

$$\begin{aligned}
\frac{\partial f}{\partial y} &= f_y(x,y) = \lim_{k\to 0} \frac{f(x,y+k) - f(x,y)}{k} && (7)\\
&= \lim_{k\to 0} \frac{-3xk + 4yk + 2k^2 + 5k}{k} && (8)\\
&= \lim_{k\to 0} \frac{k(-3x + 4y + 2k + 5)}{k} = \lim_{k\to 0}(-3x + 4y + 2k + 5) && (9)\\
&= -3x + 4y + 5 && (10)
\end{aligned}$$

DERIVATIVES OF A FUNCTION OF TWO VARIABLES

- **Example 2:**

$$\begin{aligned}\frac{\partial g}{\partial x} &= \frac{\partial}{\partial x}\left[\sin(x^2y - 2x + 4)\right] && (11)\\ &= \cos(x^2y - 2x + 4)\frac{\partial}{\partial x}[x^2y - 2x + 4] && (12)\\ &= (2xy - 2)\cos(x^2y - 2x + 4). && (13)\end{aligned}$$

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DERIVATIVES OF A FUNCTION OF TWO VARIABLES

- **Example 2 [Cont.]:**

$$\begin{aligned}\frac{\partial g}{\partial y} &= \frac{\partial}{\partial y}\left[\sin(x^2y - 2x + 4)\right] && (14)\\ &= \cos(x^2y - 2x + 4)\frac{\partial}{\partial y}[x^2y - 2x + 4] && (15)\\ &= x^2\cos(x^2y - 2x + 4). && (16)\end{aligned}$$

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FUNCTIONS OF MORE THAN TWO VARIABLES

Let $f(x, y, z)$ be a function of three variables. Then, the partial derivatives of f with respect to x, y, and z respectively written as $\partial f/\partial x, \partial f/\partial y$, and $\partial f/\partial z$ are defined to as

$$\frac{\partial f}{\partial x} = f_x(x, y, z) = \lim_{h \to 0} \frac{f(x+h, y, z) - f(x, y, z}{h}$$

$$\frac{\partial f}{\partial y} = f_y(x, y, z) = \lim_{k \to 0} \frac{f(x, y+k, z) - f(x, y, z}{k}$$

$$\frac{\partial f}{\partial z} = f_z(x, y, z) = \lim_{m \to 0} \frac{f(x, y, z+m) - f(x, y,}{m}$$

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FUNCTIONS OF MORE THAN TWO VARIABLES

We can calculate a partial derivative of a function of three variables using the same idea we used for a function of two variables. For example, if we have a function f of x, y, and z, and we wish to calculate $\partial f/\partial x$, then we treat the other two independent variables as if they are constants, then differentiate with respect to x

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FUNCTIONS OF MORE THAN TWO VARIABLES

- **Example 3:** $f(x, y, z) = x^2 - 3xy + 2y^2 - 4xz + 5yz^2 - 12x + 4y - 3z$

$$\frac{\partial f}{\partial x} = f_x(x, y, z) = 2x - 3y - 4z - 12. \tag{20}$$

$$\frac{\partial f}{\partial y} = f_y(x, y, z) = -3x + 4y + 5z^2 + 4 \tag{21}$$

$$\frac{\partial f}{\partial z} = f_z(x, y, z) = -4x + 10yz - 3 \tag{22}$$

HIGHER-ORDER PARTIAL DERIVATIVES

$$\frac{\partial^2 f}{\partial x^2} = f_{xx} = \frac{\partial}{\partial x}\left[\frac{\partial f}{\partial x}\right] \tag{23}$$

$$\frac{\partial^2 f}{\partial y \partial x} = f_{xy} = \frac{\partial}{\partial y}\left[\frac{\partial f}{\partial x}\right] \tag{24}$$

$$\frac{\partial^2 f}{\partial x \partial y} = f_{yx} = \frac{\partial}{\partial x}\left[\frac{\partial f}{\partial y}\right] \tag{25}$$

$$\frac{\partial^2 f}{\partial y^2} = f_{yy} = \frac{\partial}{\partial y}\left[\frac{\partial f}{\partial y}\right] \tag{26}$$

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HIGHER-ORDER PARTIAL DERIVATIVES

- **Example 4:** $f(x,y) = xe^{-3y} + \sin(2x-5y)$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{\partial f}{\partial x}\right] = \frac{\partial}{\partial x}\left[e^{-3y} + 2\cos(2x-5y)\right] \tag{27}$$

$$= -4\sin(2x-5y) \tag{28}$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left[\frac{\partial f}{\partial x}\right] = \frac{\partial}{\partial y}\left[e^{-3y} + 2\cos(2x-5y)\right] \tag{29}$$

$$= -3e^{-3y} + 10\sin(2x-5y) \tag{30}$$

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SELECTED APPLICATIONS

- Tangent Planes and Linear Approximations
- Gradient Vector, Tangent Planes and Normal Lines
- Relative and Absolute Minimums and Maximums
- Minimums and Maximums
- Lagrange Multipliers

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CONCLUSION

- Partial derivatives are crucial tools in multivariable calculus.
- They quantify the rate of change of a function concerning each variable independently.
- Understanding partial derivatives is essential for various applications in physics, engineering, and optimization.

LASU-VLAP

Your attendance for this lecture can not be marked at this moment because it was scheduled for Mon., Jan. 27, 2025 8:00 AM (WAT) - Fri., Jan. 31, 2025 6:00 PM (WAT) and you are outside the specified lecture period.

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